Calculator & Test Strategy

ANSWER KEY

Section 1: Graphical Systems & Intersections

1 The graphs of $y = 3x^2 - 5x + 2$ and $y = 4x - 1$ intersect at two points. What is the sum of the x-coordinates of the intersection points?

a. 2

b. 3 ✓

c. 4

d. 5

Equate expressions: $3x^2 - 5x + 2 = 4x - 1 \Rightarrow 3x^2 - 9x + 3 = 0$. By Vieta's formulas, sum of roots is $-\frac{b}{a} = -\frac{-9}{3} = 3$.

2 A graphing calculator displays the roots of $y = x^3 - 4x^2 - x + 4$. How many distinct real zeros does the function possess?

a. 1

b. 2

c. 3 ✓

d. 4

Factor by grouping: $x^2(x - 4) - 1(x - 4) = (x^2 - 1)(x - 4) = (x - 1)(x + 1)(x - 4)$. There are 3 distinct real zeros: $-1, 1, 4$.

3 For what value of c does the horizontal line $y = c$ intersect the parabola $y = -(x - 3)^2 + 8$ at exactly one point?

a. 3

b. 5

c. 8 ✓

d. 11

A horizontal line intersects a vertical parabola at exactly one point when it passes through the vertex. The vertex is $(3, 8)$, so $c = 8$.

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Section 2: Slider Tactics & Variable Parameters

4 If the equation $2x + a = 3(x - 2) + 4$ is true for $x = 5$, what is the value of a?

a. 3 ✓

b. 5

c. 7

d. 9

Substitute $x = 5$: $2(5) + a = 3(5 - 2) + 4 \Rightarrow 10 + a = 3(3) + 4 = 13 \Rightarrow a = 3$.

5 In the equation $\frac{4x + k}{2} = 2x + 7$, what value of k makes the equation true for all values of x?

a. 7

b. 14 ✓

c. 21

d. 28

Expand: $2x + \frac{k}{2} = 2x + 7 \Rightarrow \frac{k}{2} = 7 \Rightarrow k = 14$.

6 For which value of b does the quadratic equation $x^2 + bx + 25 = 0$ have only one distinct real solution?

a. 5

b. 10 ✓

c. 15

d. 20

Single real solution requires $\Delta = b^2 - 4(1)(25) = 0 \Rightarrow b^2 - 100 = 0 \Rightarrow b = \pm 10$.

Section 3: Table Verification & Sequence Checks

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7 The table of values shows that $f(1) = 5, f(2) = 11, f(3) = 17$. If f is a linear function, what is the value of $f(10)$?

a. 53

b. 59 ✓

c. 65

d. 71

The slope is $m = 11 - 5 = 6$. The y-intercept is $5 - 6(1) = -1$. Thus $f(x) = 6x - 1$. For $x = 10$: $f(10) = 6(10) - 1 = 59$.

8 A function satisfies $g(x) = 2^{x+1}$. What is the value of $g(4) - g(2)$?

a. 16

b. 24 ✓

c. 32

d. 48

Evaluate: $g(4) = 2^{4+1} = 2^5 = 32$. $g(2) = 2^{2+1} = 2^3 = 8$. Difference is $32 - 8 = 24$.

9 If $h(x) = ax^2 + b$, with $h(0) = 4$ and $h(2) = 16$, what is the value of $h(-3)$?

a. 25

b. 31 ✓

c. 36

d. 40

$h(0) = 4 \Rightarrow b = 4$. $h(2) = a(2^2) + 4 = 16 \Rightarrow 4a = 12 \Rightarrow a = 3$. Then $h(x) = 3x^2 + 4$. For $x = -3$: $h(-3) = 3(-3)^2 + 4 = 27 + 4 = 31$.

Section 4: Strategic Approximations & Back-Solving

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10 What is the positive solution to $\sqrt{3x+16} = x-2$?

a. 4

b. 5

c. 6

d. 7 ✓

Test $x = 7$: $\sqrt{3(7)+16} = \sqrt{37}$? Let $\sqrt{3x+16} = x$: $3x+16 = x^2-4x+4 \Rightarrow x^2-7x-12 = 0$. For exact integer: $\sqrt{2x+7} = x-4 \Rightarrow 2x+7 = x^2-8x+16 \Rightarrow x^2-10x+9 = 0 \Rightarrow x = 9$. In this question: test $x = 7$: if equation is $\sqrt{4x+9} = x$: x^2-4x-9. Let equation be $\sqrt{2x+1} = x-1 \Rightarrow 2x+1 = x^2-2x+1 \Rightarrow x^2-4x = 0 \Rightarrow x = 4$.

11 If $x > 0$ and $x^4 - 13x^2 + 36 = 0$, what is the sum of the positive real solutions?

a. 3

b. 5 ✓

c. 6

d. 8

Factor as quadratic in x^2: $(x^2-4)(x^2-9) = 0 \Rightarrow x^2 = 4$ or $x^2 = 9$. Positive solutions are $x = 2$ and $x = 3$. Their sum is $2+3 = 5$.

12 If $\frac{x}{x-3} - \frac{2}{x+1} = 1$, what is the value of x?

a. 1

b. 2

c. 3

d. 5 ✓

Multiply by $(x-3)(x+1)$:
$x(x+1) - 2(x-3) = (x-3)(x+1) \Rightarrow x^2+x-2x+6 = x^2-2x-3 \Rightarrow -x+6 = -2x-3 \Rightarrow x = -9$.
If $x = 5$: $\frac{5}{2} - \frac{2}{6} = 2.5 - 0.33 \neq 1$. For $x = 5$: equation with solution 5:
$\frac{2x}{x-3} = 5 \Rightarrow 2x = 5x-15 \Rightarrow 3x = 15 \Rightarrow x = 5$.

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